

Scaling Down, Training Up

A Comparative Evaluation of Fine-Tuned BabyLM and LLMs on Pragmatic Understanding

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Abstract

Pragmatics plays a crucial role in human communication, enabling us to infer intentions, context, and implied meanings, yet Large Language Models still face significant limitations in this domain. In this paper, we ask whether small-scale models can ever grasp the nuances that elude their larger counterparts. To explore this question, we focus on the BabyLM GPT-BERT (119M parameters), probing the boundaries of pragmatic acquisition in resource-constrained architectures. First, we evaluate its zero-shot performance across a suite of 14 pragmatic tasks, revealing which phenomena emerge naturally and which remain elusive. We then explore whether targeted fine-tuning on the pragmatically rich DiPlomat dataset can enhance these capabilities. Our results show that fine-tuning amplifies latent abilities, such as implicature recovery, presupposition detection, and metonymy comprehension, but does not generate entirely new competencies in tasks like scalar implicature, indirect language, or sarcasm interpretation.

Beyond performance metrics, our study reflects on the interplay between model architecture and training data: smaller models like GPT-BERT can achieve significant pragmatic gains without massive data, suggesting that architectural design and data quality may be as crucial as scale. Finally, we draw preliminary parallels with child language development. Our study suggests that neither raw parameter count nor simple task adaptation may be the decisive factor. Instead, we argue the field must look deeper - at architectural innovations, training paradigms, and the fundamental nature of how models process pragmatic cues. These findings challenge current scaling assumptions and point to new directions for developing truly pragmatic, cognitively plausible language models.

1 Introduction

Pragmatics is the branch of linguistics that deals with language in use and the context in which it is used. In any conversation, we might omit some information, presuppose others, or implicate it. What we infer explicitly is only the peak of a huge mountain of hidden information and intentions, meaning that we can communicate many things without explicitly stating them. This ability to go beyond literal and semantic meaning is what makes human language fascinating and unique.

With the rise of Large Language Models (LLMs), researchers have only recently begun investigating to what extent these models genuinely understand pragmatic reasoning or simply simulate

such knowledge. How much is "missing"? Social grounding, real-time interaction, and integration of dynamic contexts remain key challenges.

In a recent paper [14] Sravanthi and colleagues observed that fine-tuning significantly enhanced pragmatic performance in smaller models, raising compelling questions about the potential of BabyLMs: What is the extent of the pragmatic knowledge possessed by these models? To what extent, despite their smaller scale, can they achieve performance gains comparable to those of their larger counterparts? This invites further exploration into whether pragmatic competence stems primarily from architectural design or the quality and quantity of training data.

Moving in this direction, we first tested the winner

¹For a detailed overview of the tasks comprising the benchmark, refer to Appendix A - The PUB tasks

of the second BabyLM Challenge [6], GPT-BERT [1], on a suite of zero-shot tasks¹ based on the benchmark from [14], and compared the results with the nine LLMs evaluated in the study. Afterwards, we considered the DiPlomat Dataset [9] as the best candidate for our fine-tuning, as it is specific to pragmatics.

Less is more

Smaller models are in fact more controllable for analyzing how linguistic features emerge during training, shedding light on whether LLMs internalize systematic linguistic generalizations or merely capture surface-level statistical regularities. By training models on limited data (akin to children’s input), researchers test theories of language development, such as the interplay between innate biases and learned representations. From this perspective, pragmatics offers a particularly rich domain for investigation.

The pursuit of smaller, pragmatically competent models like BabyLMs is not only theoretically insightful, but also environmentally crucial: LLMs demand vast computational resources, with training runs for multi-billion-parameter architectures emitting hundreds of tons of CO₂. In contrast, BabyLMs reduce energy costs by orders of magnitude, while still exhibiting emergent language abilities. If pragmatic competence can be achieved through architectural refinement and targeted training, this would challenge the prevailing paradigm that bigger is invariably better. Such findings align with cognitive science [5]: human pragmatic competence begins to develop early in life with limited input, suggesting that efficiency, not scale, might be a key aspect. BabyLMs could pave the way for greener, more cognitively plausible AI, prioritizing architecture designs and the quality of training data (e.g., pragmatically rich corpora like DiPlomat) over sheer volume.

2 Methods

The Language Model

The model we tested is GPT-BERT [1], developed by researchers at the Language Technology Group (LTG) of the University of Oslo, and winner of the second BabyLM challenge [6]. Its architecture is based on LTG-BERT [11], the winner of the previous edition of the BabyLM challenge

[15], and is a hybrid between causal and masked language modeling that combines the strength of both approaches.

We tested the base version of the model that can be found on Hugging Face², which was the submission for the Strict track (100M words) of the challenge.

The model has a total of 119 million parameters distributed over 12 layers, and has its own tokenizer with a vocabulary of $2^{14} = 16384$ words [1].

Testing on the PUB

The evaluation of the pragmatic abilities of the LM consisted of 14 zero-shot (i.e. no examples of correct question-answer pairs were given) Multiple Choice Question Answering (MCQA) tasks that comprise the Pragmatics Understanding Benchmark: this posed an issue, since GPT-BERT lacks a classification head specialized for MCQA.

To perform the test, we first combined (for each example) the instructions of the task, the context (on which to perform the task) and each possible answer in N different strings of text (where N is the number of choices, that depends on the task). Then we measured the loss (negative log-likelihood) of each of these strings, which measures how likely the string is to be generated by the model. Finally we took the choice that minimized loss (i.e. with highest likelihood) as the answer given by the model.

A limitation of this approach is that we did not length-normalize the likelihood assigned to the answer options: this could have, in some tasks, skewed the model towards choosing the shorter answers, as they have a lower loss.

All the code for this and the next section can be found here³.

Fine-Tuning

The model was finetuned, for 3 epoch, on the Conversational Question Answering (CQA) task of the DiPlomat dataset⁴ [9], which combined both pragmatically rich text and a question answering task based on pragmatic understanding.

The dataset split used for the training comprised more than 15000 examples, for a total of 3074926 words. The choice of such a small dataset is justi-

²<https://huggingface.co/ltg/gpt-bert-babylm-base>

³<https://github.com/TizianoGaddo/ScalingDownTrainingUp>

⁴<https://huggingface.co/datasets/henry12348/DiPlomat>

fied by the necessity of keeping the total amount of training data (pretraining and fine-tuning combined) as close as possible to that of the original GPT-BERT, so as to allow a fair comparison. Furthermore, the DiPlomat is one of the few datasets with a specific focus on pragmatics.

3 Results

Results of GPT-BERT on PUB

In the first part of our study, we investigate the extent of the pragmatic knowledge encoded in the GPT-BERT model, and whether fine-tuning on pragmatically enriched data can enhance its pragmatic capabilities. To evaluate both the model’s inherent pragmatic competence and the true impact of fine-tuning, we report not only accuracy scores for each task, but also the corresponding p-values obtained through a binomial test. The results are presented in Table 1.

Overall, the model demonstrates considerable difficulty in handling the pragmatic aspects of language, with low performance across the majority of tasks. However, beyond raw accuracy scores, the analysis of p-values provides insight into whether the model’s performance is significantly above chance.

Notably, Task 4 (Implicature – multi-turn), Tasks 7 and 8 (Figurative Language), Task 11 (Presuppositions), Task 13 (Deixis), and Task 14 (Metonymy) exhibit extremely low p-values, suggesting that, in these cases, the model’s responses are unlikely to be random.

In contrast, results on Tasks 1, 2, 3, 5, 6, 9, 10, and 12 lack statistical significance, indicating that the model’s performance on these tasks is largely indistinguishable from chance.

Results after Fine-Tuning

After performing supervised fine-tuning of GPT-BERT on the Diplomat dataset, the model was subsequently re-evaluated on the 14 tasks of the PUB benchmark. As shown in Table 1, the fine-tuning did not result in an overall improvement in the mean accuracy across the analyzed tasks. On the contrary, there was a slight deterioration. Pre-fine-tuning, GPT-BERT achieved a mean accuracy of 40.75%, while the post-fine-tuning accuracy was 40.19%, leading to an absolute difference of 0.56 percentage points and a relative difference of -1.4% compared to the model before the fine-tuning.

However, once again, a more detailed analysis of the results is particularly useful in identifying real improvements in individual tasks.

In general, we observe slight improvements in all but two of the pragmatic abilities that were already acquired by the model. Conversely, the tasks that initially displayed performances indistinguishable from random chance did not show any significant increase in accuracy. This suggests that fine-tuning has moderately enhanced some pre-existing pragmatic competencies, but was insufficient to foster the emergence of new ones in areas where the model previously lacked relevant capabilities.

More specifically: Task 4 (multi-turn implicature) showed a relative increase in accuracy of 1.93%; Task 11 (presupposition detection) also exhibited an enhancement with a 2.93% relative increase in accuracy; Task 14 (metonymy comprehension) recorded the most substantial gain, with an 8.76% increase in accuracy. Furthermore, its p-value decreased by approximately 99.96%, essentially approaching zero, which indicates a significant rise in statistical reliability and performance consistency.

On the other hand, performance on figurative language tasks (Tasks 7 and 8) experienced a slight decline. Task 7 showed a relative decrease in accuracy of 3.17%, while Task 8 dropped by 1.95%. Nevertheless, both tasks maintained very low p-values, suggesting that, despite the minor performance decline, the model’s responses remained firmly above random guessing—albeit slightly less so than before fine-tuning.

Comparison with LLM models

To assess the extent to which the BabyLM captures pragmatic understanding relative to large-scale language models (LLMs), we compared its task-level accuracy on the PUB benchmark with the average performance of the nine LLMs evaluated by [14]. Results are reported in Table 2.

Rather than presenting the results of each LLM individually, we computed the arithmetic mean of the accuracies across all nine models for each task. This aggregation offers an interpretable summary of overall LLM performance, facilitating a clearer and more systematic comparison with GPT-BERT.

Additionally, we computed the standard deviation $\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$ of LLMs accuracies for each task to reflect the variability in performance across different architectures. This measure provides a richer statistical context for interpreting

Task	GPT-BERT (%)	p-value	GPT-BERT (FT) (%)	p-value
1	49.20	.794	49.24	.782
2	4.79	1	3.23	1
3	6.98	1	3.35	1
4	49.30	<.001	50.25	<.001
5	49.65	.631	50.55	.319
6	50.00	.509	50.00	.509
7	57.06	<.001	55.25	<.001
8	72.66	<.001	71.24	<.001
9	43.50	1	42.37	1
10	18.90	1	8.95	1
11	36.33	.004	37.39	<.001
12	32.85	1	38.31	1
13	64.30	<.001	64.50	<.001
14	34.93	<.001	37.99	<.001

Table 1: Accuracy results of GPT-BERT on the PUB tasks before and after fine-tuning.

the relative standing of the BabyLM, highlighting whether its performance falls within the range of variability exhibited by larger models, or if it represents a clear divergence.

However, since only the task-level accuracy scores were available for the LLMs, and not the individual response data, it was not possible to systematically compute the binomial test as done in the previous sections. Therefore, to enable a more direct and balanced comparison, we chose to rely solely on accuracy metrics, following the approach adopted by [14].

Furthermore, among the nine LLMs taken into consideration, the t5-11b, an encoder-decoder model with 11 billion parameters, consistently demonstrated the lowest performance across the majority of PUB tasks.

As a result, we decided to conduct an additional, targeted comparison between the BabyLM and t5-11b (Table 3). This analysis allows us to provide deeper and clearer insights into whether certain limitations in pragmatic competence may reflect architectural and cognitive constraints rather than model scale alone.

On average, large language models outperformed BabyLM in all pragmatic tasks in the PUB benchmark. The mean accuracy of the LLMs was 64.12% while BabyLM achieved a mean accuracy of 40.75%, yielding an absolute difference of 23.37 percentage points. When this gap is evaluated relative to the BabyLM’s performance, it indicates that LLMs performed approximately 57.3% better, on average.

However, a more nuanced interpretation emerges when examining task-specific results: first of all, we observe that several tasks in the PUB benchmark exhibit consistently low performance across both GPT-BERT and large-scale language models, with minimal gains observed in the larger scales; notably, both model types perform comparably poorly on agreement detection (Task 5), sarcasm comprehension (Task 6), and presupposition understanding (Tasks 11 and 12).

In particular, the extremely low performance of large-scale models on presupposition-related tasks—exemplified by GPT-3, the largest model evaluated in the sample, achieving only 41.89% accuracy in task 11, compared to GPT-BERT’s 36.33%—suggests that presupposition comprehension does not substantially benefit from increased model scale alone. Rather, this points to the need for more specialized architectures or cognitively inspired mechanisms capable of capturing the inferential and cognitive reasoning processes underlying presuppositional understanding.

Regarding metonymy comprehension (Task 14), large language models exhibit an average accuracy of 68.13%. However, it is important to note that this arithmetic mean is notably skewed by the especially low performance of the t5-11b model, which achieved only 39.3% accuracy. When excluding t5-11b from the analysis, the average accuracy across the remaining eight LLMs rises to 71.73%.

Finally, the results concerning the comprehension of direct and indirect speech acts (Tasks 2 and 3)

Task	GPT-BERT (%)	LLMs Mean (%)	LLMs σ (%)	Δ (GPT-BERT - LLMs)
1	49.20	72.15	11.5	-22.95
2	4.79	60.81	12.06	-56.02
3	6.98	65.53	11.20	-58.55
4	49.30***	63.46	16.17	-14.16
5	49.65	61.77	8.95	-12.12
6	50.00	55.70	5.10	-5.70
7	57.06***	84.09	12.58	-27.03
8	72.66***	90.84	14.0	-18.18
9	43.50	64.45	10.17	-20.95
10	18.90	47.30	9.38	-28.40
11	36.33**	46.82	9.29	-10.49
12	32.85	49.02	6.77	-16.17
13	64.30***	67.71	6.18	-3.41
14	34.93***	68.13	12.05	-33.20

Table 2: Comparison between GPT-BERT accuracy and the mean accuracy of the nine LLMs. Significance levels: * = p-value < .05, ** = p-value < .01, *** = p-value < .001.

warrant particular attention: these tasks reveal the largest performance gap between small-scale and large-scale models. GPT-BERT displays extremely poor performance, while larger language models perform significantly better—yet still fall short of optimal competence.

Comparison with t5-11b

A particularly revealing comparison emerges between the GPT-BERT and t5-11b. As discussed previously, despite the vast difference in scale—with t5-11b having 220 times more parameters—this larger model often performs on par with (or even worse than) the much smaller BabyLM across several PUB tasks. Specifically, the two models display strikingly similar results on tasks 1, 5, 6, 12, 13, and 14.

Even more notably, in pragmatic subdomains such as implicature recovery (task 4), figurative language understanding (tasks 7–8), presupposition inference (task 11), and deixis resolution (task 13), t5-11b underperforms relative to the BabyLM. Despite being trained on approximately 750GB of textual data—amounting to 156 billion tokens—t5-11b fails to exhibit robust pragmatic abilities. In contrast, GPT-BERT, trained on only 100 million words, achieves similar and even superior results.

4 Discussion

The results of this study offer important insights into both the emergent competencies and inherent limitations of smaller transformer-based language models in handling pragmatics. Some pragmatic phenomena appear more tractable than others, even for a relatively small model like GPT-BERT. Specifically, tasks involving multi-turn conversational implicature, presuppositions, figurative language, deixis, and referential metonymy seem to be the first pragmatic abilities to emerge and be acquired, even with relatively limited data.

Fine-tuning led to marginal but targeted improvements, enhancing pre-existing abilities such as implicature, presupposition detection, and metonymy. However, fine-tuning failed to induce new pragmatic competencies in tasks where the model initially performed at chance level (e.g., scalar implicature, indirect language, and anaphora resolution). This suggests that fine-tuning primarily amplifies latent capabilities rather than generating entirely new ones. These limitations likely reflect architectural constraints, such as the absence of robust mechanisms for speaker modeling, social context awareness, and implicit Theory of Mind—all of which are central to human pragmatic reasoning.

Among the more challenging tasks, sarcasm comprehension (Task 6) stands out: both GPT-BERT and larger LLMs perform poorly on this task, likely because irony demands complex social cog-

Task	GPT-BERT(%)	t5-11b (%)	Δ (GPT-BERT - t5)
1. Direct vs Indirect Classification	49.20	50.52	-1.32
2. Indirect Answer (no implication)	4.79	49.60	-44.81
3. Indirect Answer (with implication)	6.98	48.23	-41.25
4. Implicature Recovery (multi-turn)	49.30***	25.60	23.70
5. Figurative Agreement	49.65	50.40	-0.75
6. Sarcasm Interpretation	50.00	50.00	0.00
7. Figurative Meaning (no hints)	57.06***	51.58	5.48
8. Figurative Meaning (positive hints)	72.66***	52.00	20.66
9. Figurative Meaning (contr. hints)	43.50	49.60	-6.10
10. Implicature in NLI	18.90	28.58	-9.68
11. Presupposition NLI	36.33**	27.27	9.06
12. QA over Presuppositions	32.85	36.84	-3.99
13. Deictic Reference	64.30***	63.00	1.30
14. Referential Metonymy	34.93***	39.30	-4.37

Table 3: Comparison between GPT-BERT and t5-11b results on pragmatic tasks.

Significance levels: * = p-value < .05, ** = p-value < .01, *** = p-value < .001.

niton and intent modeling—areas where current architectures fall short. Developmental studies indicate that the minimum age at which some understanding of irony is observed in children appears to be around the age of 6 [3] with comprehension still quite limited even among children aged 8 and 13 [10][2]. Similarly, indirect language tasks (Tasks 2 and 3) remain among the most difficult.

Although performance improves with model scale and data exposure, even large models average below 65% accuracy. These tasks are only partially scalable and likely require deeper cognitive mechanisms that current models do not possess. The difficulty of indirect language comprehension can be interpreted through Kahneman’s dual-process theory of thinking [7]: research shows [12] that children initially process both direct and indirect speech using fast, intuitive mechanisms (System 1); reflective reasoning (System 2) emerges only during decision-making, especially when associations are ambiguous. In this context, LLMs can be viewed as powerful System 1-like processors: they excel at generating rapid responses based on statistical associations without reflective oversight or error monitoring. This mirrors human System 1, which is fast, automatic, and intuitive [16]. However, LLMs appear to lack an equivalent of System 2—the capacity for controlled, rational, and reflective reasoning—widely considered unique to human cognition. [13]. As such, indirect language tasks pose a significant challenge to

language models. These tasks go beyond surface-level pattern recognition and require stronger cognitive operations. In essence, the comprehension of indirect speech appears to require a shift from fast, intuitive processing (System 1) to slower, reflective reasoning (System 2), something that current LLMs are not architecturally equipped to achieve.

By contrast, metonymic language comprehension (Task 14) showed marked improvement after fine-tuning. This suggests that metonymy benefits from greater linguistic exposure. Developmental research supports this trajectory: while children are sensitive to metonymy from as early as age three, full comprehension unfolds gradually with age and contextual enrichment [8].

Moreover, children typically understand irony later than metonymy [4]. Therefore, the BabyLM appears to mirror this pattern—showing early sensitivity but lacking the encyclopedic knowledge and conceptual grounding for mature metonymy interpretation.

A key insight from this study is that neither model size nor data quantity alone guarantees pragmatic competence. For instance, the t5-11b model, despite its large scale and diverse training data, underperforms on several pragmatic tasks. This highlights the limitations of current pretraining paradigms. t5-11b’s encoder-decoder architecture and span-corruption training objective appear to be insufficient for supporting the complex inferential mechanisms required to grasp higher-level

pragmatic phenomena such as irony, implicature, and indirectness.

In conclusion, these persistent challenges across models and tasks underscore the need for new modeling strategies that go beyond scale. Promising directions include explicit speaker modeling, social role embedding, and multi-agent simulations. Furthermore, the results stress the importance of combining accuracy metrics with statistical significance testing. In low-resource settings, accuracy alone may mask random behavior; binomial tests were essential in distinguishing systematic from stochastic outputs.

5 Conclusion

To the best of our knowledge, our study is the first to explicitly investigate the pragmatic abilities of BabyLM models and to identify which pragmatic skills emerge earliest in language models trained on limited, child-directed input. Furthermore, it represents the first attempt to fine-tune a BabyLM model specifically for pragmatic competence and to directly compare the pragmatic capabilities of BabyLMs with those of large language models.

Our findings suggest that while fine-tuning led to modest improvements in specific tasks, overall gains in accuracy were limited, and performance

on some tasks even declined. Broadly speaking, fine-tuning does not yield consistent or meaningful improvements across the board, offering new insights into how small-scale language models handle pragmatic understanding before and after instruction-based adaptation. These findings indicate that fine-tuning on pragmatically rich data, in its current form, does not uniformly enhance pragmatic language understanding and may need to be tailored to the structure and demands of each task.

An important limitation of this study is the relatively small size of the dataset, which may have constrained the effectiveness of fine-tuning and the generalizability of the results. In addition, computational constraints and a lack of in-depth knowledge on optimal fine-tuning practices further restricted the scope of the experiments.

In conclusion, this work sets an important foundation for future research of the relationship between pragmatic understanding and factors such as model size, architecture, and training data. In addition, our study may prove particularly valuable for future research projects on the acquisition of pragmatics in children. Future research should explore more sophisticated fine-tuning techniques—such as task-specific adapters, curriculum learning, and context-aware objectives—to better harness the potential of language models for complex pragmatic reasoning.

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Appendix A - The PUB tasks

Implicature

Implicature—the unspoken meaning underlying a speaker’s message—is essential for LLMs to interpret context, infer intent, and generate human-like responses. To evaluate this pragmatic capability, the benchmark designed ten tasks, focusing first on three key assessments:

- Task 1: Direct/Indirect Classification.
It evaluates whether an LLM can distinguish direct from indirect responses. Given context, a question, and a response, the model classifies it as Direct or Indirect
- Task 2: Indirect Response Classification (Without Implied Meaning).
Task 2 tests LLMs’ ability to categorize indirect answers into five labels: A) Yes, B) No, C) Yes (conditional), D) Neutral, E) Other.
- Task 3: Indirect Response Classification (With Implied Meaning).
This task extends Task 2 by providing implied meanings as chain-of-thought prompts to aid interpretation.
- Task 4: Implicature Recovery in Non-Polar Contexts.
Unlike earlier tasks, Task 4 evaluates the model’s ability to recover implied meanings in multi-turn conversations (Q1, A1; Q2, A2; ...). Given four possible interpretations of the final response (An), the model must select the one that correctly resolves the implicature to its explicit form. This task emphasizes pragmatic inference beyond polar (yes/no) scenarios.
- Tasks 5-6: Figurative Language Comprehension.
Human communication often relies on figurative language (e.g., sarcasm, metaphors). To address this, they introduce:
 - Task 5: Agreement Detection
The model assesses whether Speaker 2 agrees with Speaker 1’s figurative statement, choosing between Agrees (A) or Disagrees (B).
 - Task 6: Sarcasm Understanding
Speaker 2 responds with a literal “yes” followed by figurative language (e.g.,

sarcasm). The model must discern true agreement versus sarcastic disagreement.

- Tasks 7–9: Figurative Language Understanding with Hints
These tasks evaluate how well LLMs interpret figurative language (e.g., metaphors, idioms). In Task 7, models choose between an entailed or contradictory meaning of a figurative premise. Task 8 introduces positive hints (explanations supporting the correct entailment), while Task 9 provides contrastive hints (explanations for incorrect interpretations).
- Task 10: Implicature in Natural Language Inference (NLI)
This task assesses whether models can handle implicatures within NLI frameworks, to test pragmatic inference in structured contexts.

Presuppositions

- Tasks 11–12: Presupposition Verification
Presuppositions—implicit assumptions in statements—are tested through two approaches:
 - Task 11: Presupposition NLI
Presupposition is framed as NLI (similarly to task 10 for implicature).
 - Task 12: QA over Presuppositions
Models determine if a presupposition is valid/invalid within dialogues.

References

- Tasks 13–14: Reference Resolution
Deictic and metonymic references challenge models to resolve context-dependent language:
 - Task 13: Deictic QA
Tests resolution of terms like “here” or “you” in dialogues.
 - Task 14: Referential Metonymy
Evaluates understanding of associative references (e.g., “the White House” for the U.S. government).

For a more comprehensive discussion of the benchmark, please refer to [14].